▶ MORAN CERF | *Neuroscientist*

THE AGE OF HUMANS 2.0

With billions of neurons and countless complicated neural connections, trying to understand the functioning of the human brain has always kept neuroscientists busy. But in the age of technology and AI, neuroscience is no longer just about studying the brain. It is about how we can turn our understanding of the brain functions into solutions for modern-day problems. Call it brain hacking or neural rewiring—a number of companies and individuals are keen on using neuroscience to help make better decisions and address audiences and problems more efficiently.

"The idea here is that you can take a human brain, you can put a device inside the head, and this device interfaces with the brain from within, but also with the cloud outside, and gives us the power of the Internet, of all of the information that's out there. So we can think better and become smarter," explains Moran Cerf.

While right now, much of this new science is focused on justifiable pursuits—for instance, helping those with neurological impairments regain motor skills—there is a very real probability that one day anyone could get a neural implant drilled into their brains to be able to think like a superhuman, bypassing cognitive limitations like confusion, poor memory or inability to concentrate. Elon Musk's company Neuralink is already working to develop such a product.

But there's a flip side to it, too—the prospect of our thoughts getting hacked into. According to Cerf, such a situation isn't as far away as one would think it to be. "The jump from neural implants from being a science fiction idea to reality," he says, "is now getting very, very short because people are already working on it." And what's his take on the ethical aspect of it? "It's not the first time it happened," says Cerf. "Nuclear weapons are another example...let's develop technologies that are remarkable, but let's also train people on how to use those technologies for good." ■

KEY TAKEAWAYS

- **Neuroscience is no longer about studying the brain but about using the knowledge of its functioning to find ways to become smarter and more efficient**
- **A world where we will be able to drill neural implants into our heads for better decision-making is not far away. People like Elon Musk feel it will save the human race by finding cures for diseases**
- **There are others who feel it could be the end of the human race, exposing us to hacking into our thoughts. These concerns need to be discussed now**



Photograph by BANDEEP SINGH

"OUR BRAIN HAS BEEN THE SAME BRAIN FOR MANY, MANY MILLENNIA, AND THERE ARE SOME LIMITATIONS TO WHAT THIS BRAIN CAN DO. WE HAVE A HARDER TIME DEALING WITH THE WORLD AROUND US, AND THE QUESTION IS, CAN WE ENHANCE THE BRAIN BY ESSENTIALLY CREATING A DIGITAL COMPONENT THAT WILL INTERFACE WITH THAT? THAT'S THE NEURAL IMPLANT"